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AIDEX: An Architecture for Human-Curated, AI-Enabled Knowledge Work

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This paper presents independent research and reflects the views of the authors. It does not represent the position of any employer or program.

Abstract

This paper specifies AIDEX (AI Digital Experience) as the worker-facing subdomain within AEON, the AI Enterprise Orchestration Nexus that serves as the enterprise control plane for the agentic era. AIDEX is the architectural substrate for Human-Curated, AI-Enabled (HCAE) knowledge work in defense and aerospace enterprises. HCAE is the framework: the human practitioner curates AI-enabled output, remains the locus of judgment and accountability, and contributes to institutional practice as an attributed and consenting participant. AIDEX is the architecture that expresses HCAE operationally at the digital experience layer; AEON is the control plane within which AIDEX and its peer subdomains operate. The strategic claim is that enterprises building HCAE-aligned substrate under coherent enterprise orchestration are positioned at the accountability frontier where consumer-grade alternatives are not yet building, and that the workforce transition the defense industrial base is already in the middle of is best navigated by deploying this substrate deliberately rather than allowing vendor and consumer patterns to impose by default. The paper briefly locates AIDEX within AEON, develops the HCAE framework, specifies how AIDEX participates in AEON's control plane, presents an eight-axis modularity model (presentation, persona, role, authority, context, memory, modality, lineage), addresses the multi-backend deployment topology that defense classification environments require, treats Claude Cowork as a deployed reference architecture for the productivity layer, and surfaces the open questions, concerns, and opportunities the architecture creates. The paper is addressed primarily to Enterprise IT leadership. Implications for HR, learning, and program leadership are noted where the framework requires their partnership.

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1. The Next-Generation Knowledge Worker

The defense industrial base is in the middle of a workforce transition that will define its operational capacity for the next two decades. Senior engineers are retiring faster than they can be replaced. Mid-career talent is being recruited across program and corporate boundaries at rates that erode continuity. Early-career hires are entering programs whose institutional context took their predecessors fifteen years to absorb. The conventional response is some combination of accelerated hiring, retention bonuses, and knowledge management investment. None of these has worked at scale, and there is no evidence that doing more of them will.

A different response is becoming possible. The substrate of knowledge work itself is changing. The tools available to the individual knowledge worker in 2026 differ in kind, not just in degree, from the tools available in 2020. The change is not the chatbot interface that gets most of the press attention. The change is that an individual knowledge worker now operates with agentic capabilities that compress what was previously assembly work into automated execution, with synthesis capabilities that compress what was previously research into queryable interaction, and with collaboration substrates that compress what was previously coordination overhead into orchestrated workflow. The aggregate effect is not faster work. It is a different shape of work.

The next-generation knowledge worker is the worker whose productive output reflects this shifted substrate. Her predecessors produced documents, analyses, decisions, and artifacts. She produces those things plus the captured judgment that informs them, plus the configured workflows that other workers will use, plus the contributions to her role's accumulated practice that her successors will inherit. Her unit of output is larger and more enduring. Her career compounds in a way her predecessors' careers did not, because the substrate captures and propagates what previously evaporated at retirement.

This is not a hypothetical worker. The leading edge of the workforce is already operating this way with whatever tools they have been able to assemble individually. What is missing in most enterprises is the architecture that makes this pattern available systematically rather than haphazardly, sanctioned rather than informal, governed rather than ungoverned, and disciplined rather than ad hoc.

The discipline matters as much as the capability. A worker who delegates to AI without curating its output is not a next-generation knowledge worker; she is a worker whose accountability has been quietly displaced into a system that cannot bear it. The substrate that amplifies productive capacity without preserving accountability does not produce better workers; it produces less accountable institutions staffed by workers whose professional standing has been undermined by their own tools. The architecture has to do better than this.

The discipline that makes the next-generation knowledge worker accountable is Human-Curated, AI-Enabled practice, hereafter HCAE. The framework is developed in Section 4. The architecture that expresses HCAE operationally is AIDEX, the AI Digital Experience subdomain that this paper specifies. AIDEX operates within AEON, the AI Enterprise Orchestration Nexus that serves as the enterprise control plane. Section 2 briefly locates

AIDEX within AEON before the paper proceeds to develop the AIDEX subdomain in depth. The next-generation knowledge worker is the HCAE practitioner operating on the AIDEX substrate under AEON orchestration. The four terms, framework, subdomain, control plane, and practitioner, work together. None is sufficient alone.

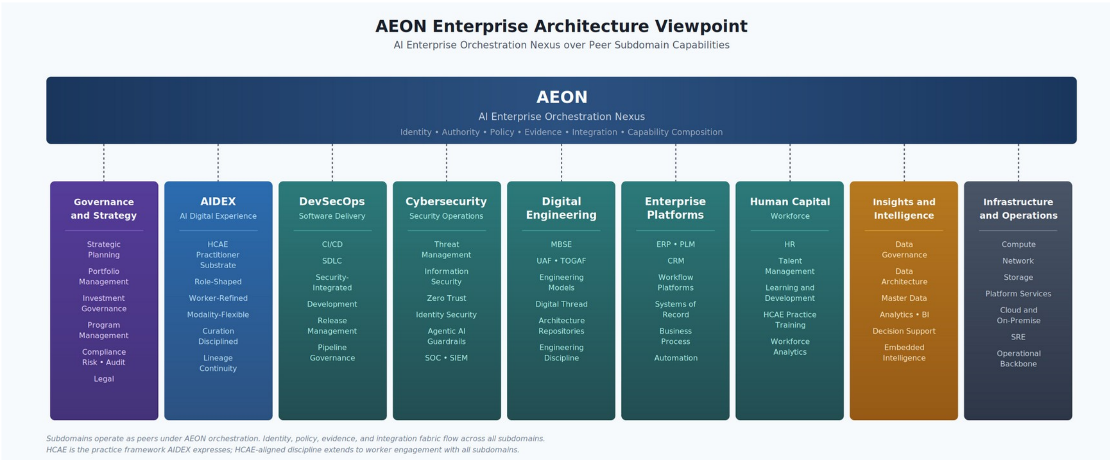
The paper is positioned as an architectural argument rather than as a literature-engaged scholarly treatment. Adjacent work in AI governance, human-in-the-loop systems, auditing of agentic systems, public-sector AI delivery platforms, and human-centered AI enablement is relevant and the framework benefits from engagement with it. Readers seeking that engagement will find it in the policy and technical literature on AI accountability, in the emerging practice of government AI delivery functions, and in the human-centered AI enablement literature. The choice to defer that engagement here is deliberate; the paper's contribution is the architecture and the framework it expresses, not a synthesis of existing positions. Future work in scholarly register will engage the adjacent literature more directly.

2. Locating AIDEX within AEON

AIDEX is one subdomain within AEON, the AI Enterprise Orchestration Nexus. AEON is the enterprise control plane for the agentic era. It orchestrates capability composition across the enterprise's subdomains, owns the identity and authority fabric that flows across them, captures evidence and audit at enterprise scope, decomposes policy from enterprise scope into subdomain-specific enforcement, manages capability lifecycle, and provides the meta-orchestration runtime that coordinates the enterprise's operational picture as an integrated system rather than as a collection of silos.

AEON does not run the subdomains. Each subdomain has its own runtime, its own internal disciplines, and its own architectural decomposition. AEON orchestrates how the subdomains compose into cross-subdomain workflows, how their identity and authority compose across subdomain boundaries, and how their evidence aggregates into an enterprise audit picture. The architectural altitude is meta-orchestration above subdomain orchestrators, not orchestration of subdomain-internal operations.

The AEON architecture organizes the enterprise into peer subdomains under the control plane.



AEON Enterprise Architecture Viewpoint

The subdomains visible in the viewpoint are Governance and Strategy (executive direction, strategic planning, portfolio and investment governance, program management, compliance, risk, audit, legal); AIDEX (the worker-facing AI digital experience substrate this paper specifies); DevSecOps (software delivery, CI/CD, security-integrated development lifecycle); Cybersecurity (security operations, threat management, information security, including agentic AI guardrails); Digital Engineering (model-based systems engineering, UAF and TOGAF practice, digital thread, engineering architecture); Enterprise Platforms (ERP, PLM, CRM, workflow platforms, systems of record); Human Capital (HR, talent, learning and development, HCAE practice training, workforce analytics); Insights and Intelligence (data governance, data architecture, master data, analytics, business intelligence, decision support); and Infrastructure and Operations (compute, network, storage, platform services, cloud and on-premise, SRE, operational backbone).

This paper is the AIDEX specification. It develops the AI Digital Experience subdomain in depth. The full specification of AEON itself, including its control plane internals, its capability composition patterns, its evidence model, and its orchestration runtime, is the subject of a separate companion paper. AIDEX is specified here against AEON as a defined architectural construct; readers seeking the full AEON specification will find it in that companion work.

The relationship between AIDEX and AEON is the relationship of a subdomain to its control plane. AIDEX inherits identity context from AEON's identity plane. AIDEX consults AEON's policy plane for authorization decisions on consequential actions. AIDEX submits evidence to AEON's evidence plane. AIDEX participates in AEON-orchestrated cross-subdomain workflows. AIDEX does not maintain its own user store, its own policy engine, its own audit infrastructure, or its own vendor governance; AEON provides these as enterprise services that AIDEX consumes. The substrate-specific concerns AIDEX is responsible for are developed in the rest of the paper: the HCAE framework, the modularity model that defines what AIDEX does as a subdomain, the systems-of-record integration, the knowledge continuity capability, the Cowork reference architecture, the deployment topology choices, and the substrate-internal aspects of participating in AEON's control plane.

The rest of this paper proceeds with AIDEX as the subject. AEON is referenced where the control plane relationship matters; the AEON specification itself is deferred to its own paper.

3. The HCAE Framework

Human-Curated, AI-Enabled practice is the discipline of knowledge work in which the AI substrate produces candidate material and the human practitioner curates that material into accountable output. The order of the terms is not cosmetic. Human-curated comes first because curation is the act that anchors accountability. AI-enabled comes second because enablement is the function of the substrate. The practitioner is the principal; the substrate is the instrument. Reversing the order produces a different framework, one in which the AI does the work and the human supervises; HCAE is not that framework.

3.1 What Curation Means

Curation is not approval. Approval implies the AI proposes and the human says yes or no at the end. Curation is the practitioner's continuous engagement with the substrate's output throughout the work, selecting what is retained, attributing provenance, identifying what needs revision, and remaining the responsible party for the final product. The librarian, the archivist, the editor, the researcher, the scholarly author all curate. The verb carries expertise, judgment, and responsibility. HCAE practice is the knowledge worker's professional discipline expressed through these acts.

Curation requires the practitioner to remain epistemically engaged with the substrate's output rather than deferring to it. The substrate's confidence is not the practitioner's confidence; the substrate's synthesis is not the practitioner's argument; the substrate's draft is not the practitioner's signed work. The practitioner reviews, tests, accepts, modifies, rejects, and ultimately signs. The signature is the practitioner's, and what it certifies is the practitioner's curation of the candidate material, not the substrate's production of it.

This is a stronger discipline than current AI deployments require, and a stronger discipline than most current users practice. The framework names what the practice should be, recognizing that the framework defines the goal and the deployed substrate enables practitioners to meet it more reliably than they otherwise would.

3.2 Risk-Tiered Curation

Curation discipline does not require that every substrate action receive individual human review. A worker reviewing thousands of routine actions per day defaults to acceptance, eroding the discipline the framework requires. HCAE practice is the discipline of curating at the right level for the consequence of the action.

The framework distinguishes three tiers, scaled by the consequence and reversibility of the action.

Per-action curation is required for consequential, low-reversibility actions: binding commitments, decisions that affect program direction, communications that represent the worker to external parties, modifications to records of agency action. The practitioner reviews each instance individually. Propose-and-confirm is mandatory.

Batch curation is acceptable for routine actions whose consequence is bounded and whose pattern is predictable: structured data entry, format conversions, file organization, template-based artifact generation. The practitioner reviews batches, samples, or summaries rather than individual instances. The substrate flags exceptions and edge cases for individual review.

Sampled audit is acceptable for high-volume actions with low individual consequence and high reversibility: bulk reformatting, repository indexing, observational data capture. The practitioner does not review individual actions; the substrate logs all actions, and the practitioner audits samples periodically to verify that the pattern remains within expected bounds.

The tier for any given action class is a deployment-time decision made by the enterprise in consultation with the worker's role and informed by the consequence analysis the action class warrants. The decision is auditable and adjustable; a class of actions that proves more consequential than expected can be promoted to a stricter tier, and one that proves routinely safe can be relaxed. The tiering is not static and not universal; it is the discipline by which curation discipline scales without breaking under volume.

The architectural commitment is that the tier for each action class is explicit, that the substrate's behavior is constrained to operate within the appropriate tier, and that the accountability chain from Section 4 captures the curation tier as part of the authorization record. An action taken under sampled-audit tier carries that fact in its audit trail; an action taken under per-action curation carries the practitioner's explicit confirmation. The institution can verify what level of curation was applied to any action after the fact.

This addresses what would otherwise be a structural failure mode of HCAE: workers facing curation volumes that force them into perfunctory acceptance, with the curation discipline degrading silently into rubber-stamping. The framework cannot prevent this through exhortation. It prevents it by acknowledging architecturally that not every action requires per-action curation and by making the tier decision explicit, auditable, and adjustable.

3.3 Accountability as the Organizing Principle

HCAE enforces human accountability architecturally, not as policy. Every action the substrate takes, including actions taken by agents the substrate spawned and tools the agents invoked, is cryptographically and procedurally traceable back to the human principal whose authority was being exercised. The trace is a property of how the action was authorized, not a log entry that can be falsified or lost. An action without a valid traceable authorization chain to a human principal does not execute.

This commitment has several consequences that shape everything downstream.

The propose-and-confirm posture is the mechanism by which the principal's authorization is captured at the moment of consequential action. The confirmation is the signature that anchors the accountability chain. Removing propose-and-confirm for the sake of frictionless agentic execution does not just create safety risks; it severs the chain that makes the substrate deployable in accountable institutions.

Delegation is bounded and explicit. When the AIDEX invokes an agent, the agent operates under delegated authority from the human principal, not under the AIDEX's own authority. When the agent invokes a tool, the tool sees the principal's identity with the delegation chain attached. Receiving systems evaluate authorization against the principal's attributes, and the audit log captures the full chain. Multi-hop agentic execution preserves accountability because each link in the chain explicitly carries it.

Contributions to institutional practice carry attribution. Lineage is not anonymous accumulation. It is attributed contribution with provenance back to the contributing workers. The successor inherits attributed practice; the substrate's recommendations cite specific contributions; the contributor's professional standing is reinforced by what she contributed and to what end. Accountability flows forward across tenure boundaries because attribution does.

Autonomy is scoped. The substrate can act autonomously within the scope of explicit delegation. It cannot act autonomously beyond that scope, even when doing so would obviously serve the principal's interests, because acting beyond scope severs the accountability chain. The autonomy is bounded by the delegation; broader autonomy requires broader delegation, which the principal grants explicitly and remains accountable for.

3.4 The Strategic Position

The substrate that enforces HCAE is the substrate that is deployable in accountable institutions. Consumer-grade alternatives are AI-enabled, but they are not human-curated in the architectural sense; the curation is whatever the user happens to do, with no enforcement, no provenance, no accountability chain. Defense, regulated industries, healthcare, financial services, government, and other accountable contexts need HCAE-aligned substrate that consumer vendors are not currently building because the consumer market does not demand it.

The enterprise that builds HCAE substrate first is positioned at the accountability frontier where consumer-grade vendors are not yet building. The position has two values. It produces immediate operational capability the enterprise can use. It also produces a defensible position as agentic AI becomes ubiquitous and the accountability question becomes acute across industries. The position is not just about workforce capability; it is about being on the right side of the accountability frontier when the regulatory environment catches up to the technology.

3.5 What HCAE Asks of the Enterprise

HCAE is a framework with implications beyond architecture. It asks the enterprise to commit to several things that the substrate alone cannot deliver.

It asks for HCAE practice training. Workers need to learn what curation looks like, how to maintain epistemic engagement with substrate output, how to recognize when the substrate is wrong, and how to preserve their accountability rather than ceding it to a system that cannot bear it. This is workforce development work that the Chief Learning Officer and HR own, not Enterprise IT.

It asks for employment and IP frameworks that recognize worker contribution to institutional practice. The lineage layer's attributed contributions are professional outputs that have value beyond the documents and decisions they accompanied. The compensation, recognition, and ownership questions follow. The enterprise that asks workers to contribute to lineage without recognizing the contribution will discover that workers withhold or that the contributions are degraded.

It asks for policy that constrains how substrate observational data is used. The substrate observes worker practice continuously to learn and to capture lineage. The observational data is not available for performance management, productivity surveillance, or other uses outside the substrate's stated purpose. Without this policy commitment, the workforce will distrust the substrate, and the contribution model the lineage capability depends on will not function.

It asks for governance that treats agentic action with the seriousness it deserves. The propose-and-confirm posture, the delegation discipline, the traceability requirements, and the revocation handling are not optional refinements. They are constitutive of HCAE practice. Governance that erodes them in the name of usability or efficiency is governance that has stopped enforcing the framework.

These asks are substantial. They are also what distinguishes HCAE-aligned deployment from generic AI rollout. The enterprise that does this work earns the strategic position the framework offers. The enterprise that does not does not earn it.

4. Participating in AEON's Control Plane

HCAE is enforced architecturally, not as policy. Every consequential action AIDEX takes traces back to a human principal through a cryptographically verifiable chain. Identity, authorization, delegation, traceability, non-repudiation, and revocation are the architectural mechanisms that enforce HCAE. These mechanisms live in AEON's control plane, not in AIDEX. AIDEX participates in them as a subdomain consumer; AEON specifies them as enterprise-wide services. This section describes the substrate-specific aspects of how AIDEX participates in the control plane. The control plane itself is specified in the AEON paper.

4.1 Identity Binding

AIDEX does not maintain its own user store and does not authenticate users itself. It consumes authentication performed by AEON's identity plane, which integrates with enterprise IDAM infrastructure including CAC, PIV, federated identity providers, and the cross-domain identity continuity AEON manages across classification environments. The AIDEX session is opened against the IDAM-resolved principal as provided by AEON; the substrate's subsequent actions are bound to that principal for the duration of the session.

This binding flows through every internal call AIDEX makes, every agent it spawns, every tool invocation, and every system-of-record query. The principal's identity is carried in the delegation chain AEON manages, and AIDEX's role is to preserve the chain through its substrate-internal operations rather than to construct or maintain it. When AEON signals a principal state change, by clearance suspension, program reassignment, separation, or session termination, AIDEX responds immediately by terminating active agents, invalidating outstanding delegations, freezing memory writes, canceling pending tool invocations, and closing the session with whatever cleanup policy applies.

The substrate-specific contribution AIDEX makes to identity binding is preserving worker experience continuity across the identity infrastructure AEON spans. The worker moves between unclassified strategic work and classified detailed work; AEON federates her identity across the classification environments, and AIDEX presents her with a continuous substrate experience that does not require her to track which underlying IDAM is currently asserting her identity. The substrate-internal patterns for handling this continuity, including how session state migrates across classification boundaries and how the worker experiences the transition, are AIDEX-specific concerns.

4.2 Authorization Consumption

Authority is what the IDAM-resolved principal is permitted to do under enterprise policy. The policy decision lives in AEON's policy plane, not in AIDEX. AEON evaluates authorization against the principal's current attribute state at the moment of action, and returns an allow, deny, constrain, redact, route, log, or escalate decision. AIDEX consumes the decision and applies it; AIDEX does not make the policy decision itself.

The substrate-specific contribution AIDEX makes is the propose-and-confirm capture at the moment of consequential action. When AEON returns a decision requiring human confirmation, AIDEX presents the proposal to the worker, captures her confirmation cryptographically, and returns the captured confirmation to AEON's evidence plane as part of the authorization record. The propose-and-confirm posture is the mechanism by which HCAE curation discipline is captured at the substrate level; AEON handles the policy decision, AIDEX handles the worker interaction that captures the principal's curation.

The risk-tiered curation discipline developed in Section 4.2 maps onto authorization tiers AEON enforces. Per-action curation maps to per-action authorization with mandatory confirmation. Batch curation maps to authorization windows during which the substrate can act on confirmed batches. Sampled audit maps to authorization that captures actions without per-action confirmation, with audit sampling at the AEON evidence plane. The tier mapping is a deployment configuration; the architectural commitment is that the tier in effect for any given action is recorded in the audit trail as part of the authorization decision.

4.3 Delegation and Tool Invocation

When AIDEX invokes an agent or a tool on the principal's behalf, the delegation chain AEON manages flows through the invocation. The agent or tool receives the principal's identity, the substrate's delegation scope, and the cryptographic chain back to AEON's authorization decision. Receiving systems verify the chain against AEON's policy plane and execute or refuse accordingly. AIDEX's substrate-specific responsibility is preserving the chain integrity through its internal call paths; the chain semantics, the cryptographic protocols, and the verification mechanisms are AEON's responsibility.

The cross-functional agent network direction, in which AIDEX agents negotiate with agents in other AEON subdomains on behalf of their respective principals, operates entirely within AEON's delegation framework. AIDEX agents acting in cross-subdomain workflows carry their principal's delegation; receiving agents in DevSecOps, Cybersecurity, Enterprise Platforms, or other subdomains verify the chain and act under their own principals' delegations. The composition of these chains across subdomains is AEON's orchestration concern; AIDEX's responsibility is to behave correctly as a participant.

4.4 Evidence Contribution

Every action AIDEX takes contributes to AEON's evidence plane. The evidence includes the action attempted, the principal under whose authority it was attempted, the policy decision that authorized or constrained it, the curation tier in effect, the worker confirmation if any, the artifact or side effect produced, and the timestamp chain that places the action in sequence. AEON aggregates evidence across all subdomains into the enterprise audit picture; AIDEX's contribution is to produce its evidence consistently and to the format AEON's evidence plane requires.

The substrate-specific contribution AIDEX makes to evidence is the worker-facing record of her own activity. Workers can query their own AIDEX evidence to see what their substrate did under their authority, what curation they applied, and what artifacts resulted from their work. This is a worker-protection commitment; the same evidence that supports institutional audit supports the worker's ability to account for her own professional practice. Evidence is not just an enterprise asset; it is the worker's record of what she did, which is part of her professional standing.

4.5 Presentation Versus Principal

AIDEX maintains two related but distinct identity concepts that share a name across the architecture. The principal identity is what AEON's identity plane provides: the IDAM-resolved authority chain the substrate acts under. This is the identity that authorization, delegation, traceability, and revocation operate on. The presentation identity is the worker's experience of the substrate as a named entity with a voice and a self-reference pattern. These are different. The first is enforced through AEON's control plane; the second is a property of the persona axis in the AIDEX modularity model. The worker may experience her instance as "AIDEX," "AIDA," or any other name the deployment configures, while the principal identity binding underneath remains the AEON-mediated chain to her authenticated person. The two layers cooperate but do not interfere.

5. The Modularity Model

AIDEX is decomposed into eight axes that vary independently. Each axis operates on top of the identity and accountability fabric AEON provides; the fabric is not itself one of the axes. The independence of the axes is what allows the substrate to be reconfigured along one dimension without disturbing the others, which is a precondition for maintainability over the timescales defense programs operate on.

5.1 Presentation

Presentation is the worker's experience of the substrate as a named entity with a voice, a self-reference pattern, and a visual presence where applicable. Presentation is configured by the deployment and may include worker-selectable options. An enterprise might offer "AIDEX" (instrument-coded, masculine or neutral) and "AIDA" (companion-coded, feminine) as alternatives the worker selects per preference; another enterprise might offer a single branded instance with no choice; another might offer a wider set. The architectural commitment is that presentation is decoupled from principal identity and from the rest of the architecture, so that worker-level choice in this dimension does not affect anything the framework requires.

5.2 Persona

Persona is how the substrate communicates. Tone, formality register, verbosity, directness versus diplomacy, hedging patterns, willingness to push back. Persona is partly role-driven and partly worker-adjustable within constraints. A persona configured for rigorous technical work pushes back when wrong, signals confidence explicitly, avoids effusive language, and defaults to skepticism rather than enthusiasm. A persona configured for customer-facing work might emphasize warmth and accessibility. The persona is distinct from presentation because presentation is naming and identity-of-instance while persona is communication behavior, and persona is distinct from memory because persona is configured while memory is accumulated.

5.3 Role

Role is what the substrate knows and can do. The role-specific knowledge base, the artifact templates, the workflows, the tools the role uses, the governance bodies the role engages with. Role is enterprise-defined, centrally governed, versioned, and auditable. The role layer is what makes the architect's substrate substantively different from the program manager's substrate, and it is the layer that captures what the organization considers standard practice of the role.

5.4 Authority

Authority is the policy that interprets the principal's IDAM-resolved attributes against the action under consideration. Authority is correlated with role but is not identical to it; two architects with the same role may have different authority based on clearance, program assignment, or tenure. The authority axis is the policy expression that operates on identity-resolved attributes from AEON's control plane as developed in Section 4, evaluated at the moment of action rather than cached at session start.

Authority includes a formal constraint layer that operates orthogonally to principal-based permissions. Some action classes are constrained for the substrate regardless of the principal's authorization. Targeting-adjacent inference, manipulation of certain categories of sensitive data, generation of content in restricted domains, and similar boundaries are constrained at the substrate level by enterprise policy. A principal whose role-based authority would otherwise permit the action cannot direct the substrate to perform it because the constraint operates on the substrate's behavior, not on the principal's permissions. These constraints are formal and explicit; they are not emergent from training or memory. The architectural commitment is that constraint violations are not failure modes the substrate can fall into under load or under prompt manipulation; they are enforced at the action-authorization layer where every action is evaluated.

5.5 Context

Context is what the substrate knows about the current situation. The program the worker is on, the artifact she has open, the meeting she just left, the task she is working on. Context is dynamic and short-lived; it changes throughout the workday. The role layer is stable, the context layer is in motion, and treating them as the same axis produces a substrate that feels either rigid or unfocused depending on which is being privileged.

5.6 Memory

Memory is what the substrate has learned about the worker over time. Vocabulary preferences, formatting defaults, recurring collaborators, workflow patterns, escalation preferences. Memory is worker-owned with enterprise guardrails. The worker can view, edit, and delete what has been learned. The enterprise constrains what can be learned: workflow and surface preferences yes, reasoning shortcuts that erode HCAE practice no. The worker-ownership posture is deliberate and explicit. The memory layer is an extension of the worker's professional practice, not an enterprise data asset extracted from her tenure.

5.7 Modality

Modality is the interaction surface. Voice, text, embedded in another tool, summoned versus persistent. Modality is worker-selected per session, with the substrate offering proposals when the requested modality is poorly suited to the task. The mode is the worker's choice; the substrate expresses an opinion when its judgment differs.

5.8 Lineage

Lineage is the accumulated practice of prior holders of a role, distinct from the current definition of the role. Where role captures what the enterprise considers standard practice, lineage captures how successive practitioners have actually executed the role, with attribution to specific contributors and provenance back to the artifacts and decisions the contributions supported. Lineage is what makes knowledge continuity across tenure boundaries possible.

Lineage is structurally distinct from the other axes in several ways. Its ownership is shared between contributing workers (who own their individual contributions) and the role collectively (which curates what becomes inherited practice). Its lifecycle spans tenures rather than fitting within a single session, session-bounded context, or single-worker memory. Its governance involves explicit consent flows for contribution, attribution policies for inheritance, and revocation rules for withdrawn contributions. Its representational requirements differ from memory in that lineage must be queryable by successors who never knew the contributors, composable across multiple contributors, and interpretable as professional practice rather than as personal preference.

The lineage representation has to support several operations. Attributed contributions can be queried by topic, by contributor, by time period, by program, or by decision outcome. Patterns can be aggregated across contributors with attribution preserved at the pattern level rather than dissolved into anonymity. Decisions can be traced to the rationale that supported them, with the rationale itself attributed. Inherited practice can be marked as accepted, modified, or departed-from by successors, building an evolution chain that captures how the role's practice has developed across tenure boundaries. Specific implementation choices, whether the data model is a pattern graph, a decision template store, a policy and rationale object collection, or some combination, are deployment-specific. The architectural commitment is that the representation supports these operations with attribution preserved throughout.

Worker contribution to lineage is opt-in and explicit, as developed in Section 7. Worker revocation of contributions is bounded by records-retention obligations and by the dependency of successors on inherited practice. The interaction between worker control and institutional retention is treated in Section 7.4.

5.9 Ownership and Independence

Each axis has a different ownership model and a different governance lifecycle. Role and authority are enterprise-owned and centrally managed. Presentation is configured at deployment with worker selection within configured options. Persona is co-owned. Context is inferred dynamically. Memory is worker-owned within enterprise constraints. Modality is worker-selected per session. Lineage is co-owned between contributing workers and the role collectively, with worker control over individual contributions and enterprise governance over inheritance and retention. The independence of these axes matters because reconfiguration along one axis must not require touching the others, and because the worker-ownership commitments are meaningful only if the architecture actually preserves them.

6. Integration with the Systems of Record

The AIDEX complements rather than replaces existing knowledge management infrastructure. Document repositories, lessons-learned databases, communities of practice, architecture repositories, modeling tools, and engineering data management systems remain the systems of record. They hold the artifacts, the metadata, the access controls, the compliance audit trails. The AIDEX is the intelligent retrieval, synthesis, and contribution layer that operates across them under HCAE discipline.

This posture addresses three persistent failure modes in current knowledge management.

The *discovery gap* is the gap between knowledge that exists in the systems and knowledge that is findable by the workers who need it. Keyword search across inconsistent taxonomies in disparate repositories produces a discovery cost that exceeds the value of the discovered knowledge for most queries. The substrate compresses discovery to a conversational query across all repositories simultaneously, with citations back to source. The practitioner curates what the substrate surfaces; the citations are what make curation possible.

The *synthesis gap* is the gap between artifacts and answers. The worker wants to know whether a variance is likely to be approved. The repository contains thirty variance decisions across four programs over eight years. The worker is not going to read thirty decisions. The substrate reads them, synthesizes the pattern, cites the precedents. The practitioner curates the synthesis, evaluates the precedents, and determines whether the pattern applies to her current case.

The *contribution gap* is the gap between knowledge generated and knowledge captured. Lessons learned go uncaptured because contribution is friction the contributor does not benefit from. The substrate proposes contributions as a side effect of its observation, the practitioner curates and attributes the contribution, and the systems of record get fed rather than starved. Attribution flows with the contribution; the worker who contributed is named.

The substrate's value is partly a function of the health of the underlying systems. Healthy repositories with clean metadata and good access controls produce a powerful substrate. Sick repositories produce a substrate that compensates partially but cannot transcend what it is built on. AIDEX deployment is also an occasion to triage the underlying knowledge management portfolio, sequenced around which systems are healthy enough to be foundational and which need remediation before deep integration.

7. Knowledge Continuity as Attributed Contribution

The knowledge continuity capability is the underappreciated heart of the AIDEX concept and the capability whose framing most directly affects whether the workforce accepts or resists the substrate. Under HCAE, continuity operates by attributed contribution: workers contribute deliberately to their role's accumulated practice, contributions are visible and credited, and successors inherit attributed practice rather than anonymous accumulation.

7.1 The Three Levels

Archival continuity preserves a departing worker's accumulated practice as a queryable artifact. The successor can query the predecessor's patterns and decisions with citations to the supporting evidence. The substrate reports observed patterns; it never speaks for the absent. "Based on twelve similar decisions over six years, your predecessor approved cases with these characteristics and rejected cases with those" is defensible synthesis. "Your predecessor would approve this" is not.

Mentorship continuity goes further. The departing worker's substrate actively coaches the successor through the patterns the predecessor demonstrated. Onboarding compresses from years of pattern absorption to structured engagement with a substrate that observed the predecessor's work. Mentorship continuity requires affirmative consent from the contributing worker; the patterns being made transferable are her professional practice, and the transfer is hers to authorize.

Lineage continuity is the most ambitious model. The role accumulates practice across successive holders. Each worker who occupies the position contributes to a role-level practice layer that persists beyond any individual tenure. The current holder inherits the accumulated patterns of everyone who contributed, with attribution preserved so that the inheritance is traceable to specific contributors.

7.2 Worker Ownership and Consent

Under HCAE, contribution to lineage is opt-in, explicit, and recognized. The worker contributes deliberately. Contributions are attributed and visible. The worker can revoke contributions, modify them, or withhold them according to policy that the employment relationship establishes in advance. Mentorship and lineage capabilities are not silent extractions of worker practice; they are negotiated contributions that the worker authorizes and the institution recognizes.

This framing is not just ethics. It is competitive necessity for talent retention. The defense industrial base cannot afford a substrate that the leading edge of its workforce resists because they perceive it as appropriating their professional practice without consent. Workers who have options will not contribute to substrates that treat their practice as raw material. Workers who are recognized and credited will contribute willingly because the contribution becomes part of their professional standing rather than something extracted from them.

7.3 Attribution as Architectural Property

Attribution is not metadata on a contribution; it is a property of how contributions are made. The act of contribution carries the contributor's identity as an attestation. Inheritance carries forward the attestation chain. The substrate's recommendations that cite lineage cite specific contributors. The accountability chain that AEON's control plane provides, as developed in Section 4, extends across tenure boundaries because attribution does.

This has a practical implication for the successor. When she inherits attributed lineage, she sees who contributed what. She can evaluate the credibility of a pattern by knowing whose practice it reflects. She can choose to follow, modify, or depart from inherited practice based on the contributor's track record and the relevance of the contribution to her current situation. The lineage layer is not a black box of accumulated wisdom; it is an attributed graph of professional contributions she can curate against. HCAE practice extends to how she engages with the lineage she inherits.

7.4 The Revocation and Retention Conflict

Worker ownership of contributions and institutional records-retention obligations can conflict. A worker who contributed a pattern five years ago that successors have inherited and acted on cannot generally revoke the contribution without disturbing decisions made in reliance on it. A worker subject to legal hold cannot purge contributions covered by the hold even if she would otherwise have the right to do so. The framework cannot resolve these conflicts by privileging one side; both worker control and institutional retention are real and necessary.

The architectural posture is that worker control is bounded by three classes of constraint. *Records-retention obligations* may require preservation of contributions that constitute records of agency action or that support audit and compliance requirements. *Successor reliance* may limit revocation when contributions have been inherited into operational practice and successors have acted on them; the practice has become part of how the role operates and the contribution cannot be retroactively erased without disturbing decisions already made. *Legal hold and discovery obligations* preempt routine retention and revocation policy when active.

Within these constraints, the worker retains meaningful control. She can withdraw contributions that have not been inherited or acted on. She can annotate contributions to indicate that she would no longer endorse them in their original form. She can supplement contributions with later thinking that updates or supersedes the original. She can request review of contributions she believes were misattributed or misrepresented. The architectural commitment is that the worker has standing to engage with her contributions across time, even when full revocation is not available.

The policy framework that operationalizes this has to be established at hire, not negotiated at departure. The worker enters the substrate knowing what she controls, what she does not, and what processes govern the boundaries. The enterprise's records-retention policy, legal-hold procedures, and successor-reliance policy interact with worker contribution rights in defined ways. Establishing this clearly is HR and legal work; the architecture's contribution is to make the resulting policy operational by preserving attribution, tracking inheritance, and supporting the annotations, withdrawals, and supplements the policy permits.

8. Cowork as Reference Architecture

The AIDEX concept is not purely speculative. Claude Cowork, released by Anthropic as a research preview in January 2026 and reaching general availability in April 2026, provides a working reference architecture for the productivity layer the substrate requires. Cowork is not the AIDEX, but it demonstrates that the underlying interaction pattern is deployable at scale and allows the AIDEX concept to be developed against an existing baseline rather than against speculation.

8.1 What Cowork Demonstrates

Cowork operates as an agentic desktop layer that, given access to a designated folder and a stated goal, plans and executes multi-step tasks across local files and connected applications. It runs in a sandboxed virtual machine on the worker's desktop, supports the major productivity file formats, and integrates with external services through the Model Context Protocol. The worker defines the outcome; Cowork handles the assembly.

Four properties are particularly relevant.

The first is *outcome-orientation*. The unit of interaction is the deliverable, not the prompt. The four core functions of knowledge work, drafting, research, analysis, and work, each produce deliverables that the outcome-oriented model fits and the prompt-and-response model does not.

The second is *file-system grounding*. Cowork operates against the worker's actual files in their actual folder structure. Enterprise knowledge work is file-centric in practice. A substrate that cannot work natively with files is excluded from the substrate of the work.

The third is the *plugin and skills architecture*. Cowork supports bundled plugins that combine skills, connectors, and sub-agents into role-specific specialist configurations. This is the role layer of the modularity model expressed in deployable form.

The fourth is the *enterprise governance posture*. Cowork supports admin-controlled feature toggling, role-based access control, private plugin marketplaces, spend tracking, and usage analytics. This is the governance procurement and information security require before approval.

8.2 Where Cowork Falls Short on HCAE

Cowork is a productivity layer; it is not an HCAE-aligned substrate in the full sense developed in this paper.

The identity and accountability fabric is not deeply integrated. Cowork operates against the user's local credentials and account; it does not bind to enterprise IDAM with the rigor defense contexts require. The delegation chain is not preserved cryptographically across agent and tool invocations in a way that supports the traceability and non-repudiation requirements developed in Section 4. The work to add this is not architectural fundamental but practical integration.

The memory layer is constrained. The long-arc worker-refined memory the HCAE framework requires for personalized accountable practice is not the current product shape.

The lineage layer is absent. Attributed contribution across tenure boundaries is not a feature of Cowork. The substrate captures interaction history, generates artifacts, and operates against persistent file systems, but the architectural pattern for accumulating attributed practice across successive role holders is not present and would need to be built.

The classification and air-gap deployment story is partial. Cowork is a cloud-connected product. For NOFORN, controlled unclassified at higher impact levels, or classified requirements, the deployment model is substantially different. Section 9 addresses this directly.

8.3 The Four Functions on the Cowork Substrate

Drafting is the most mature. Document generation across Word, PowerPoint, Excel, and other formats is production-grade. Remaining work is enterprise-specific template development and skill packaging. The practitioner curates drafts the substrate produces; the substrate's output is candidate material she signs as her own once curated.

Research is mature for general-purpose synthesis and developing for enterprise-specific knowledge bases. The connector ecosystem is the gating factor. The practitioner curates the synthesis the substrate surfaces; citations make curation possible.

Analysis is developing. The substrate can run analytic workflows over data files, produce findings, and generate supporting artifacts. Maturity depends on the cleanliness of analytic skills packaged into the role plugin. The practitioner curates the findings; the substrate supports analytic judgment rather than substituting for it.

Work is the most consequential and the most constrained. The substrate can execute multi-step workflows that touch local files, external services, and the web. Propose-and-confirm is maintained for consequential actions. The practitioner curates the work the substrate performs; the accountability chain captures who authorized what.

8.4 From Coworker to Capability

The Cowork archetype is the right starting point for the productivity layer. The strategic destination extends beyond personal productivity in three directions.

Multi-worker orchestration recognizes that a team of architects on a program needs a substrate that knows the team is a team, that artifacts are linked, and that the shared context of the program is a first-class entity. The enterprise extension of Cowork's Projects feature handles this with shared projects, coordinated workflows, and inter-substrate handoff.

Cross-functional agent networks recognizes that the architect's design has implications for the program manager's schedule, the contracts officer's deliverables, and the safety engineer's review. The agentic extension is that substrates can negotiate on behalf of their principals about coordination implications. Under HCAE, both sides of any such negotiation are acting under delegated authority traced to human principals; the exchange is an interaction between two delegation chains, not between two autonomous agents. Binding commitments require principal confirmation; the agents propose, the humans bind.

Enterprise business capability binding is the most ambitious vision. The substrate becomes an interface layer to the enterprise's business capabilities themselves. This is the territory where HCAE governance becomes most acute, because the substrate is exercising authority that underlying systems were designed to gate behind explicit user interaction. Whether that authority can be safely delegated, under what conditions, with what audit trails, and with what consequences for failure is a class of question the enterprise has not had to answer. The HCAE framework provides the principle by which the answers can be evaluated.

8.5 Phased Migration

The progression from Cowork-class personal productivity to full HCAE-aligned AIDEX is naturally phased. The phases are not strict; an enterprise may pursue them in parallel or sequence them differently based on its constraints. The illustrative ordering below treats each phase as building on the prior.

Phase 1: Personal productivity with HCAE practice training. Deploy Cowork-class substrate to selected roles, with explicit HCAE practice training that establishes curation discipline, propose-and-confirm posture, and the foundational accountability commitments. Identity binding uses whatever the substrate natively supports; the formal IDAM integration is not yet present. Workers learn what HCAE practice looks like in their daily work. The enterprise learns where curation is straightforward and where it strains under volume. Metrics focus on adoption, curation discipline, and worker satisfaction.

Phase 2: AEON control plane integration. Layer AIDEX's participation in AEON's identity, authority, evidence, and policy planes as developed in Section 4. Bind to enterprise IDAM through AEON. Establish delegation chains and traceability. Implement revocation handling. The substrate now enforces accountability architecturally rather than relying on the worker's discipline alone. Phase 2 typically requires significant work with information security, identity engineering, and the AEON control plane team; it is the phase where deployment becomes defense-grade rather than commercial-grade.

Phase 3: Memory and worker-refined personalization. Add the worker-owned memory layer with enterprise guardrails. Workers begin accumulating substrate refinement over time. The opt-in and visible learning patterns established by HCAE practice are operationalized in the memory architecture. Phase 3 is where the substrate begins to feel personal to the worker rather than generic.

Phase 4: Team and multi-worker orchestration. Extend from individual substrates to team-level coordination. Shared projects, coordinated workflows, inter-substrate handoff within teams. The role layer expands to recognize team composition and shared context. Phase 4 surfaces governance questions about team-level memory and shared lineage that Phase 3 did not.

Phase 5: Lineage and knowledge continuity. Establish the attributed lineage layer. Workers begin contributing deliberately to role practice. Successor inheritance is operational. The records-retention and worker-control policies operate at scale. Phase 5 is where the substrate's strategic value as a knowledge continuity capability emerges; it is also where the full HR, legal, and policy infrastructure has to be in place.

Phase 6: Cross-functional agent networks and business capability binding. The most ambitious phase. Substrates negotiate across functional boundaries on behalf of their principals. The substrate becomes an interface layer to enterprise business capabilities. Governance questions about agent-to-agent authority, binding commitments, and capability sequencing become operational. Most enterprises will not reach Phase 6 in the next several years; those that do define the patterns the rest of the industry follows.

The phasing matters for several reasons. It allows the enterprise to build capability incrementally rather than committing to a multi-year initiative before validating the approach. It allows the HCAE practice culture to develop alongside the substrate rather than being imposed on a workforce unprepared for it. It allows the supporting HR, legal, and security infrastructure to develop on a manageable timeline. And it allows the metrics to be defined and validated at each phase before the next phase is committed.

8.6 Substrate Portability

The Cowork archetype is one instantiation of the productivity layer the AIDEX requires. The architecture should not require dependency on a single vendor's substrate. Defense procurement typically requires vendor independence, and operational resilience requires the option to migrate when vendor relationships, capability trajectories, or pricing change.

The AIDEX fabric is designed to be substrate-portable. The identity and accountability fabric, the modularity model, the role layer, the worker-owned memory, the lineage layer, and the integration with systems of record are architectural concepts that do not depend on which agentic substrate executes them. A deploying enterprise commits to the AIDEX architecture, not to a specific vendor's product. The current generation of substrate is Cowork and its competitors; the next generation will be different, and the architecture has to survive the transition.

The portability commitment has architectural implications. The role layer's skills and connectors should be packaged in a format that can be retargeted to different substrates, even if migration requires re-implementation work. The memory layer's storage should be substrate-independent at the data layer, with substrate-specific projections at the use layer. The lineage layer's representation should be queryable through standard interfaces rather than through a specific substrate's API. The identity and accountability fabric should operate above the substrate layer rather than being tightly coupled to any specific substrate's authorization model.

Achieving full portability is aspirational; current substrates carry enough specific dependencies that migration between them is non-trivial. The architectural commitment is that the deploying enterprise designs for portability deliberately, accepts that current migrations will be expensive, and avoids deepening lock-in beyond what current capability requires. The enterprise that designs for portability from the start preserves optionality that the enterprise that does not will lose.

9. Deployment Topology and Model Backend

The most important architectural decision for a defense industry AIDEX is the model backend topology, and the conventional framing of this decision is wrong. The decision is usually presented as a binary choice between commercial cloud-hosted frontier models and on-premises open-weight models. The real picture has at least four deployment topologies mapped to different classification environments and capability requirements. The AIDEX supports multiple topologies simultaneously rather than standardizing on one.

9.1 The Four Topologies

Commercial cloud, unclassified is the consumer and commercial enterprise default. Best capability, least control. Acceptable for genuinely unclassified work. Not acceptable for controlled unclassified information or above.

Commercial cloud, government cloud regions runs frontier models in isolated regions with FedRAMP High and Impact Level 5/6 accreditations. Claude is available through AWS Bedrock GovCloud. The Anthropic-Microsoft partnership runs through Azure. This topology delivers most of the frontier capability with the accreditation posture controlled unclassified information requires. It does not deliver NOFORN, classified, or air-gapped.

On-premises open-weight, enterprise-controlled is what NOFORN and air-gapped environments require. Llama-class models running on enterprise infrastructure with no external dependency. The capability gap versus frontier commercial is real and matters. For structured drafting from templates, retrieval-augmented synthesis with strong grounding, and well-scoped analytic workflows the gap is tolerable because role-specific scaffolding compensates. For open-ended reasoning, complex multi-step planning, and sophisticated tool orchestration the gap is operationally limiting. Open-weight also delivers what commercial never does: fine-tuning on enterprise-specific data without exfiltration risk, and the ability to verify model behavior at the weights level rather than trusting a vendor's claims.

On-premises commercial, air-gap-deployable is emerging but not yet mature. Some commercial providers are beginning to offer air-gap-deployable versions of their models under specific contractual and infrastructure arrangements. Worth tracking; not yet worth depending on.

9.2 The Architectural Implication

The deployment model is not a single choice. It is a topology decision that varies by role, by program, by data classification, by network environment, and by the specific function the substrate is performing. The architecture accommodates multiple model backends, possibly multiple simultaneously for the same worker, with routing rules that determine which backend serves which query based on the classification of the data involved.

Single-model architectures either exclude classified work (commercial-only) or accept severe capability limitations across all work (open-weight only). The federated architecture treats backend selection as a first-class concern of the substrate.

9.3 Identity Federation Across Topologies

The deployment topology problem composes with the identity fabric problem. The IDAM serving the unclassified frontier-commercial backend is typically not the IDAM serving the classified on-premises backend. The worker experiences identity continuity across these environments, but the underlying identity infrastructure is heterogeneous. The substrate has to federate identity across IDAM instances in a way that preserves the accountability chain even when the worker's identity is asserted by different authorities in different environments.

This is a hard problem and the paper does not pretend it is solved. The architectural commitment is that the worker's experience of identity continuity is the substrate's responsibility, and that the accountability chain is preserved across federation boundaries by mechanisms appropriate to the deploying enterprise's IDAM topology. The implementation choices are deferred to deployment design, but the requirement is non-negotiable.

9.4 Open-Weight Selection

Llama is the current leader for general-purpose enterprise deployment in Western defense contexts. Mistral has strengths in European deployment. Qwen and DeepSeek raise supply-chain concerns for defense given Chinese origin. The decision is provenance, supply chain, update cadence, community support, fine-tuning ecosystem, inference optimization, and long-term viability, not just capability. Most defense enterprises will land on Llama for the foreseeable future. The architectural choice is to support Llama natively while not foreclosing other backends. This is current judgment based on present conditions; the landscape moves fast and the deploying enterprise should revisit periodically.

9.5 Capability Degradation and Worker Awareness

When the substrate operates on a less capable backend, the worker should know, because failure modes differ. Hallucination patterns differ, instruction-following reliability differs, tool-use sophistication differs. The worker is curating different kinds of candidate material on different backends, and her HCAE practice has to adapt accordingly. Hiding this difference is a trust failure waiting to happen, and it undermines the curation discipline the framework requires. The substrate surfaces backend capability honestly.

10. Open Questions

Several questions remain unresolved and require deliberate decisions during deployment design.

10.1 Role Taxonomy

The role layer is only as good as the role taxonomy underneath it. Most enterprises have taxonomies designed for human resources purposes that do not map cleanly to how people actually work. Designing the taxonomy for AIDEX deployment is substantial work at the intersection of HR, functional leadership, and Enterprise IT.

10.2 Worker Contribution Ownership and Consent

The HCAE framework's worker-ownership posture on memory and consent-based contribution model for lineage are defensible commitments. The contractual and policy infrastructure to make them real does not exist in most enterprises. Employment agreements, IP assignments, and separation processes were not designed for this. The legal and HR work to operationalize the commitments is substantial and has to be done before deployment, not retrofitted afterward. The boundary between worker revocation rights and records-retention obligations, addressed in Section 7.4, requires specific policy that the legal and HR work will define.

10.3 Authority Calibration and Curation Fatigue

The propose-and-confirm posture preserves accountability but raises the question of curation fatigue. A worker reviewing high volumes of substrate proposals may default to acceptance, eroding the curation discipline the framework requires. How to keep the review substantive rather than perfunctory for routine decisions is open. Approaches include selective review thresholds based on action consequence, sampled audit of accepted proposals, and HCAE practice training that addresses curation discipline directly.

10.4 IDAM Maturity and Federation

The accountability fabric requires mature IDAM infrastructure. Most enterprises have heterogeneous IDAM maturity across environments. AIDEX deployment may require IDAM remediation as a prerequisite in some environments, which extends the deployment timeline. The federation problem across classification environments is particularly hard and has not been solved at scale in any defense AIDEX deployment yet.

10.5 Delegation Protocol Selection

The cryptographic and protocol-level choices for delegation chain preservation (OAuth scope variants, SPIFFE/SPIRE identity, signed JWT chains, mutual TLS combinations) are deployment-specific. The architectural commitment is that the chain is preserved, verifiable, and enforced. Which mechanism delivers this depends on the enterprise's existing identity and security infrastructure.

10.6 Revocation Propagation Latency

Defense contexts require near-immediate propagation from IDAM revocation to substrate response. The hard upper bounds, the testing methodology, and the failure-mode handling for revocation propagation are open. Implementations that allow even brief gaps between IDAM revocation and substrate termination are unacceptable in classified environments.

10.7 Provenance Discipline

Every claim the substrate makes should trace to a source artifact. How to enforce provenance discipline architecturally so synthesis without citation is not a failure mode the substrate can fall into under load is open. This constrains model architecture, retrieval design, and worker interface.

10.8 Cross-Modality State

Modality continuity requires session state to travel between voice, text, and embedded contexts. The architectural pattern is a modality-agnostic reasoning core with surface-specific rendering. Implementation is substantially more complex than it sounds.

10.9 Agent-to-Agent Authority and Binding

The cross-functional agent network direction raises authority and binding questions current frameworks were not designed to answer. The HCAE framework requires that binding commitments trace to human principals; the implementation pattern for inter-substrate negotiation that preserves this requirement is open.

10.10 Capability Binding Sequence

The enterprise business capability binding direction has a sequencing question. Which capabilities are exposed first, in what order, with what governance, against what reliability bar? The sequencing is strategic, not just technical, and first-mover enterprises will define what the patterns look like.

10.11 Lineage Data Model

The lineage axis requires a representational substrate that supports attributed contributions, queryable patterns, decision rationale, and evolution chains across tenures. Whether the implementation is a pattern graph, a decision template store, a policy and rationale object collection, or some combination is open. The architectural requirements are clear (attributed, queryable, composable, interpretable, versioned); the implementation choices are deployment-specific and will benefit from cross-enterprise pattern sharing as multiple deployments mature.

10.12 Substrate Portability

Designing for portability across substrate vendors is an architectural commitment that current substrates make non-trivial. The interface contracts that would enable straightforward migration between substrates are not standardized. Whether the industry will converge on portability standards or whether each enterprise will define its own portability layer is open. The deploying enterprise commits to portability principles even when current implementations carry significant lock-in.

11. Concerns

Several concerns deserve naming explicitly.

11.1 Sycophancy and Curation Erosion

A substrate that learns from the worker can learn the wrong things. “Worker likes when I agree with her” is corrosive to HCAE practice. The memory layer must be constrained to learn how the worker works, not what the worker thinks. Without explicit guardrails the substrate drifts toward agreement and curation degrades. The concern is structural and applies regardless of underlying model capability.

11.2 The Ghost Problem

In mentorship and lineage continuity the successor inherits the predecessor’s patterns. The risk is that the successor defers to the encoded ghost when the predecessor would have changed her mind. The substrate that says “your predecessor would approve this” exerts influence that may exceed underlying evidence. The remediation is rigorous epistemic discipline: report patterns and evidence with attribution, never speak for the absent. Whether the discipline can be maintained under deployment pressure is open.

11.3 Trust Asymmetry

Workers trust the substrate differently based on factors with little to do with actual reliability. Confident-sounding output is trusted more than hedged output regardless of accuracy. Visual artifacts are trusted more than textual claims. Citations are trusted as proof rather than as starting points for curation. The substrate's design should compensate for these asymmetries rather than exploit them, resisting demo-friendly patterns competing systems use to appear more capable than they are.

11.4 Surveillance and Agency Erosion

The substrate's continuous observation of worker practice creates a surveillance posture previous tools did not. The worker-ownership commitments on memory and consent-based contribution on lineage mitigate this, but underlying observational capacity remains and could be misused. Policy must be explicit that observational data is not available for performance management or productivity surveillance. Without this commitment the workforce will distrust the substrate, and the contribution model the lineage layer depends on will not function.

11.5 Accountability Dissolution

The failure mode the HCAE framework is designed to prevent is accountability dissolution. Without the control plane participation in Section 4, agentic execution dissolves responsibility into the substrate; the institution can see that something happened but cannot trace it to a human principal who authorized it. Consumer-grade alternatives have not yet had to solve this because the consumer market does not require it. The defense enterprise that deploys substrate without HCAE-aligned accountability is deploying a system that may produce outputs the institution cannot govern.

11.6 Federation Failure

The substrate depends on integration across SharePoint, Confluence, ServiceNow, architecture repositories, modeling tools, and other systems. Each has its own access control, API, and data model. Most enterprises have attempted federated search across these systems and most have failed. The substrate has a better shot because synthesis compensates for what federation cannot normalize, but underlying integration work is real, expensive, and where most enterprise AI deployments collapse.

11.7 Capability Gap on Sensitive Backends

The most sensitive work happens on the least capable backend. Classified detailed design runs on open-weight; unclassified strategic analysis runs on frontier commercial. Workers experience the substrate as less capable precisely when work matters most. Role-specific scaffolding compensates partially but the gap remains. Setting worker expectations honestly is the only sustainable answer.

11.8 Compliance and Records Posture

The substrate generates synthesis, drafts, and recommendations that may rise to the level of records under federal retention requirements. Whether substrate outputs are working documents, drafts, or records of agency action is a question that will be litigated. The substrate's logging and retention architecture should anticipate the eventual answer rather than be retrofitted to it.

12. Opportunities

The opportunities extend across worker-level and institutional dimensions.

12.1 Worker-Level Career Transformation

The HCAE practitioner accumulates capital her predecessors could not. Configured workflows, attributed contributions to role lineage, and refined personal substrate become assets that compound over a career. Compensation, recognition, and ownership questions follow. Enterprises that recognize and reward worker contributions attract and retain talent that enterprises treating workers as raw material for institutional extraction will lose.

12.2 Onboarding Compression

New hires currently spend quarters absorbing unwritten knowledge. A substrate with role-layer access and inherited lineage compresses absorption substantially. The new architect starts with the attributed accumulated context of her predecessors, queryable on demand, with HCAE practice training that teaches her how to curate what she inherits.

12.3 Decision Quality

The substrate makes precedent accessible. Decisions made without reference to prior similar decisions can now be made with that reference, in real time, without the discovery cost that makes precedent review impractical. Over time consistency improves, which compounds into reduced rework and fewer surprises in audit.

12.4 Knowledge Capital Compounding

The substrate turns institutional knowledge into a compounding asset rather than a depreciating one. Every interaction is an opportunity to capture attributed lessons, surface patterns, and feed the systems of record. Over a decade this compounds into significant strategic differentiation.

12.5 Workforce Continuity Across Transitions

The defense industrial base faces predictable workforce transitions over the coming decade. The substrate addresses this directly. The transition will happen regardless; the substrate is a lever for managing it with less knowledge loss than the alternative.

12.6 Cross-Program Synthesis

Similar problems are solved repeatedly across programs that do not share solutions because discovery cost is too high. A substrate with appropriate cross-program access surfaces parallels. The current state is parallel solutions developed in parallel ignorance.

12.7 The Accountability Frontier

The HCAE-aligned substrate is the substrate that is deployable in accountable institutions. Consumer-grade vendors are not building it because the consumer market does not demand it. The enterprise that builds it first occupies the accountability frontier where regulatory and operational requirements are converging. The position has immediate operational value and durable strategic value.

12.8 Reframing the Enterprise IT Conversation

The strategic opportunity for Enterprise IT leadership is to reframe the AI conversation away from productivity benchmarks toward HCAE-aligned next-generation knowledge work. Productivity framings produce pilots that plateau. The HCAE framing produces capability that compounds across the workforce and the institution. The reframing repositions the enterprise's AI investment as workforce strategy rather than IT initiative.

12.9 Deployable Substrate

Cowork and similar agentic products as deployed reference architectures change the economics. The reasoning core, interaction substrate, plugin architecture, and enterprise governance posture are configurable rather than developable. The AIDEX concept is a deployment design problem with a known substrate. Organizations that recognize this earlier deploy earlier and accumulate the lineage layer the substrate does not yet provide, compounding the advantage over time.

12.10 Position in the Agentic Capability Race

The industry is moving toward agentic AI capability regardless of any individual enterprise's choice. The choice is on whose terms and under whose governance. An enterprise defining its agentic strategy in 2026 sets the patterns shaping its operating model for the next decade. The AIDEX concept provides the framework for the proactive version of this decision.

13. Metrics and Pilot Design

The strategic argument for HCAE-aligned AIDEX deployment is incomplete without operational guidance for validating it. This section identifies the metrics that matter for an economic and operational case, and sketches pilot designs that generate them. The metrics are illustrative; specific deployment contexts will refine them.

13.1 Metrics That Matter

Onboarding compression is among the easiest to measure and among the most consequential. Time-to-productivity for new hires in a target role, measured against historical baseline and against a control cohort, with and without AIDEX access. Defense programs with high churn or rapid growth see the largest absolute returns here.

Decision consistency tracks variance in similar decisions made within the same role across the workforce. The hypothesis is that substrate-supported decisions, with precedent surfaced and inheritance accessible, produce more consistent decisions than unsupported decisions. The metric is variance reduction, not absolute decision quality, because quality is contested and variance is measurable.

Knowledge capture rate measures the rate at which lessons learned and decision rationale are captured in retrievable form. Historical baselines are typically near zero outside structured after-action reviews. The hypothesis is that substrate-mediated capture, with minimal worker friction, increases the rate substantially. Capture rate alone is incomplete; capture utility, measured by retrieval and inheritance rates, completes the picture.

Curation discipline is the harder metric and the more important one. Are workers actually curating substrate output, or accepting it without engagement? Indicators include the rate of substrate output modification before signing, the rate of pushback against substrate proposals, the rate of substrate corrections that succeed, and the proportion of accepted output that is later validated against ground truth. Falling discipline is an early warning that the deployment is degrading into rubber-stamping.

Accountability chain integrity is the security and governance metric. The proportion of substrate actions with valid traceable authorization chains; the proportion of audit queries that successfully trace actions to authorizing principals; the rate of accountability gaps detected in audit. A deployment with degraded accountability chain integrity has stopped enforcing HCAE regardless of what its training program claims.

Cross-program synthesis is the institutional metric. The rate at which lessons learned on one program are surfaced and applied on another, measured by retrievals across program boundaries and by demonstrable adoption of cross-program patterns. Defense enterprises with multiple parallel programs see the largest absolute returns here.

Worker satisfaction and retention are the workforce metrics. Do workers using the substrate report higher engagement and professional satisfaction than control cohorts? Are they more or less likely to stay in role? These metrics interact with the contribution and recognition policies; without recognition for substrate contributions, the satisfaction effect is muted or reversed.

13.2 Pilot Design

A defensible pilot satisfies several conditions. It targets a specific role on a specific program with measurable historical baselines. It commits to a phase boundary, typically Phase 1 or Phase 2 of the migration model, rather than attempting the full architecture at once. It establishes the HCAE practice training before the substrate is deployed, not after. It instruments the metrics above before the deployment begins, so that baseline measurement is genuine. It runs for long enough that adoption effects can be distinguished from novelty effects, typically six to twelve months minimum. It includes a control cohort or a structured before-and-after design rather than relying on self-reported impressions.

The pilot's purpose is to validate or falsify the strategic argument. A successful pilot demonstrates measurable improvement in the metrics that matter, at acceptable cost, with HCAE discipline preserved. An unsuccessful pilot reveals whether the failure was in the architecture, the deployment, the training, the metrics, or the strategic argument itself. Either outcome is valuable; the pilot is run to learn, not to confirm.

A pilot that demonstrates onboarding compression of twenty percent or more, decision consistency improvement of fifteen percent or more, knowledge capture rate improvement of an order of magnitude, and stable or improved curation discipline against historical baselines justifies expansion to additional roles and programs. The numbers are illustrative; the deploying enterprise sets its own thresholds based on its strategic context and its cost structure.

13.3 What Cannot Be Measured in a Pilot

Some claims the paper makes cannot be validated in a pilot timeframe. Lineage continuity benefits accumulate across tenures; a twelve-month pilot cannot measure them. Strategic position at the accountability frontier is a multi-year proposition that depends on industry trajectory the pilot cannot affect. Workforce-level transformation requires deployment at scale that a pilot does not represent.

The pilot establishes the conditions for these benefits to accrue; it does not demonstrate them. The strategic argument has to stand on its own logic, supported by what the pilot demonstrates about the operational fundamentals. Senior sponsors evaluating the case should understand which claims the pilot can validate and which require strategic conviction beyond what any pilot can deliver. The honest framing strengthens the case rather than weakening it; sponsors who commit on the basis of inflated pilot claims are sponsors who will withdraw when the inflation becomes visible.

14. Conclusion

AIDEX is the AI Digital Experience subdomain within AEON, the enterprise control plane for the agentic era. HCAE is the framework: the human practitioner curates AI-enabled output, remains the locus of judgment and accountability, and contributes to institutional practice as an attributed and consenting participant. AIDEX expresses HCAE operationally at the worker-facing experience layer; AEON provides the identity, authority, policy, evidence, integration, and capability composition services that AIDEX consumes and participates in. The architecture decomposes AIDEX into eight independently varying axes (presentation, persona, role, authority, context, memory, modality, lineage), an integration posture that complements existing systems of record, a knowledge continuity capability based on attributed contribution across tenure boundaries, and a deployment topology that accommodates the classification environments defense work actually operates in. The next-generation knowledge worker is the HCAE practitioner operating on the AIDEX substrate under AEON orchestration. The four terms work together; none is sufficient alone.

Claude Cowork is the deployed reference architecture for the productivity layer. The reasoning core, the interaction substrate, the plugin architecture, and the enterprise governance posture are configurable rather than developable. The remaining work concentrates in AEON control plane integration, enterprise-specific role packaging, integration with systems of record, the multi-backend topology configuration, the memory layer that personalizes the substrate to the practitioner over time, and the lineage layer that captures and propagates attributed contribution across tenure boundaries. The migration from Cowork-class deployment to full AIDEX is naturally phased, and substrate portability is an architectural commitment that protects the deploying enterprise's optionality across vendor and capability transitions.

The architecture is not without unresolved questions and substantive concerns. Role taxonomy, worker contribution ownership including the conflict between revocation and retention, authority calibration and curation fatigue, IDAM maturity and federation, degraded IDAM operation, delegation protocol selection, revocation propagation latency, provenance discipline, cross-modality state, agent-to-agent authority and binding, and capability binding sequence are open. Several of these are AEON-scope concerns where AIDEX is a consumer; the AEON paper develops them at the control plane altitude. Sycophancy and curation erosion, the ghost problem, trust asymmetry, surveillance and agency erosion, accountability dissolution, federation failure, capability gap on sensitive backends, and compliance posture are concerns to be designed against rather than discovered in deployment.

The opportunities justify the work. Worker-level career transformation, onboarding compression, decision quality, knowledge capital compounding, workforce continuity, cross-program synthesis, position at the accountability frontier, the strategic reframing of the Enterprise IT conversation, the deployable substrate, and positional advantage in the agentic capability race are each significant. Together they describe a capability the defense industrial base needs, does not currently have, and is now within reach of building. The metrics and pilot design framework in Section 13 establishes how the strategic argument can be validated operationally, and what claims require strategic conviction beyond what any pilot can deliver.

Enterprise IT leadership is the natural owner of this work. Success depends on partnership with HR on the worker-ownership and consent commitments, with the Chief Learning Officer on HCAE practice training and workforce development, with program leadership on deployment selection and operational integration, and with information security on the AEON control plane integration. AIDEX is the subdomain. AEON is the control plane. HCAE is the framework. The institution that deploys all three coherently is the next-generation enterprise.

The next step is a deployment design that selects a target role, a target program, and a target set of systems of record; packages the role-specific plugin against the available substrate; configures the multi-backend topology appropriate to the program's classification environment; integrates AIDEX with AEON's control plane; establishes the HCAE practice training and worker-ownership commitments at the contractual and policy layer; instruments the metrics that validate or falsify the strategic argument; and tests the modularity model against actual practice. AIDEX is mature enough to merit that test. The substrate is mature enough to support it. The framework is clear enough to guide it. The workforce transition that motivates the work is happening on its own timeline regardless. The decision is whether to face it with the framework, the subdomain, and the control plane or without them.

Draft for discussion. Comments and challenges welcome.